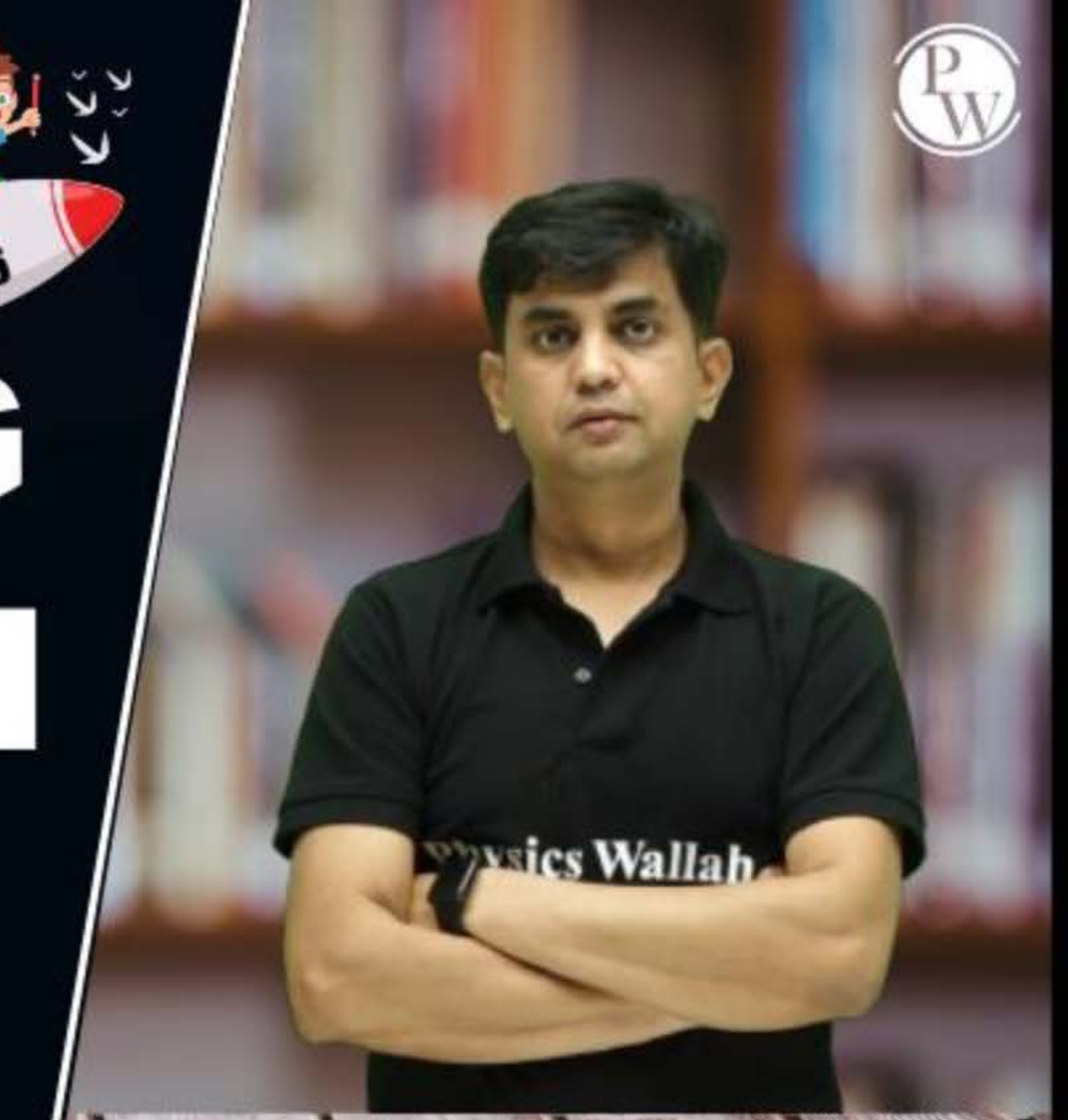
CS & IT ENGINEERING

Computer Network

Transport Layer

Lecture No. - 12



By - Abhishek Sir

Recap of Previous Lecture

Topic TCP Window Scaling

Topic TCP Timers

Topics to be Covered

Topic Socket Programming

[Network Prog.]

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW

Topic : Sockets

→ Socket is data structure inside the program

→ Both client and server programs communicate via a pair of sockets

Socket API :

→ Set of library function calls

→ Between Application Layer and Transport Layer

→ Door between application process and end-to-end transport protocol

Topic : Socket Programming

→ Client / server application that communicate using sockets

Process : Controlled by application developer

OSI Model : Mostly controlled by operating system

Topic : Types of Sockets

Stream Socket : SOCK_STREAM

- → Uses TCP protocol
- → Connection-oriented service
- → Reliable and byte stream oriented

Datagram Socket : SOCK_DGRAM

- → Uses UDP protocol
- → Connection-less service
- → Unreliable datagram

Topic : Creating a Socket

```
#include <sys/types.h>
#include <sys/socket.h>

int socket ( int domain, int type, int protocol );
```

domain : Protocol Family
[PF_INET, AF_INET, PF_UNIX etc]

type : Communication Protocol semantics
[SOCK_STREAM, SOCK_DGRAM]

protocol : Specify particular protocol, set "zero" as default

return : On success, a file descriptor for the new socket (sock_fd)
On error, -1 is returned

```
sockfd
  = socket( );
```

→ Both TCP client and server create Socket

Topic : Bind a name to a Socket

```
#include <sys/types.h>
#include <sys/socket.h>

int bind ( int sockfd, const struct sockaddr *addr, socklen_t addrlen );
```

sockfd : file descriptor of the socket

return : On success, zero
On error, -1 is returned

→ Bind : optional for TCP client.

Topic : Listen for connections

→ only by TCP server

→ TCP server is ready to listen client req.

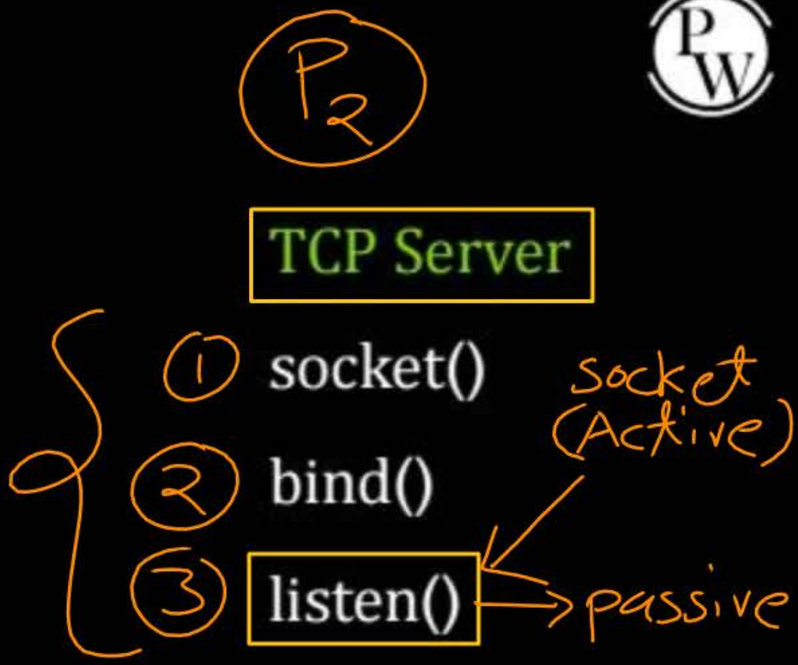
```
#include <sys/socket.h>

int listen ( int sockfd, int backlog );
```

sockfd : file descriptor of the socket at server TCP

backlog : maximum length for queue of pending connections

return : On success, zero
On error, -1 is returned

Topic : Socket Programming with TCP



#Q. Which one of the following socket API functions converts an unconnected active TCP socket into a passive socket?

[GATE-2014, Set-2, 1-Mark]

IIT-KGP

(A) connect

(B) bind

(C) listen

(D) ~~connect~~ accept

Ans: C

Topic : Initiate a connection

→ only by TCP client

```
#include <sys/socket.h>

int connect ( int sockfd, const struct sockaddr *addr, socklen_t addrlen );
```

sockfd : file descriptor of the socket at client TCP

return : If connection succeeds, zero
On error, -1 is returned

Topic : Socket Programming with TCP

P1

TCP Client

1. socket()
2. connect()

P2

TCP Server

socket()

bind()

listen()

#Q. Which of the following system calls results in the sending of SYN packets?

[GATE-2008]

11 Sec

(A) socket

(B) bind

(C) listen

(D) connect

Ans: D

Topic : Accept a connection

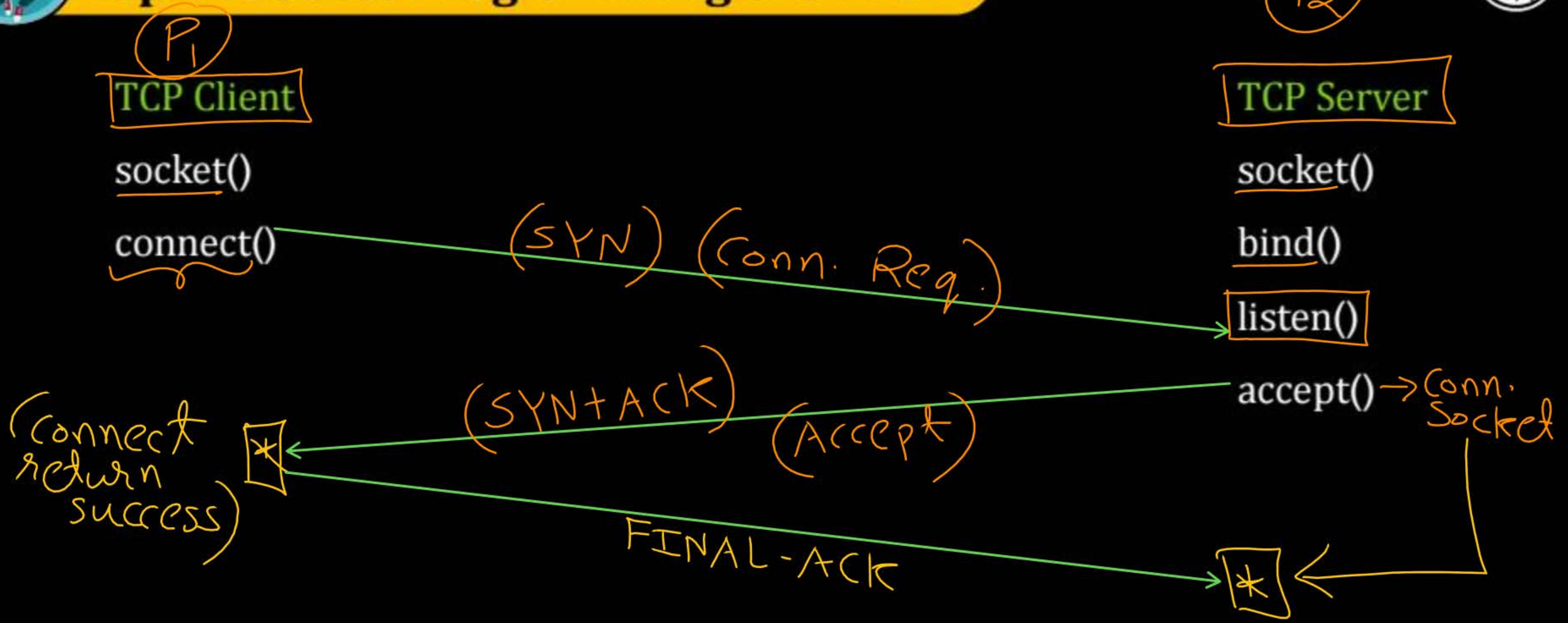
→ Creates a new connected (connection) socket

```
#include <sys/socket.h>

int accept ( int sockfd, struct sockaddr *_Nullable restrict addr,
                         socklen_t *_Nullable restrict addrlen );
```

sockfd : file descriptor of the socket (welcome socket) at TCP server

return : On success, file descriptor for the accepted socket (connection socket)
On error, -1 is returned

Topic : Socket Programming with TCP



#Q. Identify the correct order in which a server process must invoke the function calls accept, bind, listen, and recv according to UNIX socket API.

[GATE-2015, Set-2, 1-Mark]

IIT-K

(A) listen, accept, bind, recv

(B) bind, listen, accept, recv

(C) bind, accept, listen, recv

(D) accept, listen, bind, recv

Ans: B

TCP Server

1. Socket
2. Bind
3. Listen
4. Accept

#Q. A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a socket(), a bind() and a listen() system call in that order, following which it is preempted. Subsequently, the client process P executes a socket() system call followed by connect() system call to connect to the server process S. The server process has not executed any accept() system call. Which one of the following events could take place?

[GATE-2008]

IISC; H.W.

(A) connect() system call returns successfully

(B) connect() system call blocks

(C) connect() system call returns an error

(D) connect() system call results in a core dump

Topic : Write to a socket

```
#include <unistd.h>

ssize_t write ( int sockfd, const void buf[.count], size_t count);
```

sockfd : file descriptor of the socket

return : On success, the number of bytes written is returned
On error, -1 is returned

Topic : Read from a socket

```
#include <unistd.h>

ssize_t read ( int sockfd, void buf[.count], size_t count);
```

sockfd : file descriptor of the socket

return : On success, the number of bytes read is returned
On error, -1 is returned

Topic : Socket Programming with TCP

P1

P2

TCP Client		TCP Server
1 socket()		socket() 1
2 connect()	SYN →	bind() 2
		listen() 3
	← SYN+ACK	accept()
	ACK →	
write()	→	read()
read()	←	write()

Topic : Close a socket

```
#include <unistd.h>

ssize_t close ( int sockfd );
```

→ send FIN segment

sockfd : file descriptor of the socket

return : On success, returns zero
On error, -1 is returned

Topic : Socket Programming with TCP

TCP Client

socket()

connect()

TCP Server

socket()

bind()

listen()

accept()

write()

read()

close()

4-way handshake

read()

write()

close()

Topic : Server Type

→ Several clients can request service of the server in the same time.

Non-concurrent Server :

- → Service one client at one time
- → Other client request must wait

Concurrent Server :

- → Services all client requests simultaneously

Topic : Socket Programming with TCP

Non-concurrent TCP Server :-

```
socket()
bind()

while(1) {
        listen()
        accept()
        read() / write()   }

close()
```

Topic : Socket Programming with TCP

Concurrent TCP Server :-

```
socket()
bind()
while(1) {
        listen()
        if ( fork() == 0 ) {
                accept()
                read() / write()
                close()    }   }

close()
```

Topic : Send a message on socket

```
#include <sys/socket.h>

ssize_t sendto ( int socket, const void *message, size_t length, int flags,
                 const struct sockaddr *dest_addr, socklen_t dest_len );
```

sockfd : file descriptor of the socket

return : On success, the number of bytes send is returned
On error, -1 is returned

Topic : Receive a message from a socket

```
#include <sys/socket.h>

ssize_t recvfrom ( int socket, void *buffer, size_t length, int flags,
                   struct sockaddr *address, socklen_t address_len );
```

sockfd : file descriptor of the socket

return : On success, the number of bytes receive is returned
On error, -1 is returned

Topic : Socket Programming with UDP

P1

UDP Client

1. socket()
   sendto()
   recvfrom()
   close()

P2

UDP Server

1. socket()
2. bind()
   recvfrom()
   sendto()
   close()

Topic : Connected UDP sockets

P1

UDP Client

socket()

connect()

write()

read()

close()

connect

P2

UDP Server

socket()

Bind()

⟹ seperate Socket

read()

write()

close()

#Q. Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which connect function has already been called. Which of the following statements is/are CORRECT?

I. A connected UDP socket can be used to communicate with multiple peers simultaneously.

II. A process can successfully call connect function again for an already connected UDP socket.

[GATE-2017, Set-2]

IIT-R, H.W.

(A) I only

(B) II only

(C) III Both I and II

(D) Neither I nor II

2 mins Summary

Topic Socket Programming

THANK - YOU